

求积分

$$\int_0^\infty \ln \left(\frac{1 + \theta \cos x}{1 - \theta \cos x} \right) \frac{1}{\cos x} \mathrm{d}x$$

其中 $\theta \in (-1, 1)$

令

$$\begin{aligned} f(\theta) &= \int_0^\infty \ln \left(\frac{1 + \theta \cos x}{1 - \theta \cos x} \right) \frac{1}{\cos x} \mathrm{d}x \\ \Rightarrow f'(\theta) &= \int_0^\infty \frac{\mathrm{d}x}{1 - \theta^2 \cos^2 x} \\ &= \frac{\pi}{\sqrt{1 + \theta^2}} \end{aligned}$$

从而有

$$f(\theta) = \pi \ln(\sqrt{\theta^2 + 1} + \theta)$$